

SECTION B

9 (**long**) The regular tetrahedron $ABCD$ has its vertex A at the origin, vertex B at $(1, 0, 0)$, vertex C in the plane $z = 0$ with $y > 0$, and vertex D in $z > 0$.

(a) Show that the coordinates of C and D are

$$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right) \quad \text{and} \quad \left(\frac{1}{2}, \frac{1}{2\sqrt{3}}, \sqrt{\frac{2}{3}}\right)$$

respectively. [10]

(b) Determine the equation of:

(i) the plane containing BCD ; [7]

(ii) the line through A perpendicular to the plane containing BCD ; [3]

(iii) the line through D perpendicular to the plane containing ABC . [2]

(c) Find the intersection of the lines described in b(ii) and b(iii) above. What is the physical significance of this point? [5]

(d) Find the volume of this regular tetrahedron. [3]